

Electrotonic Structure Shapes Partner-Level Synaptic Organization on Cortical Dendrites

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Abstract

How synaptic inputs are spatially organized on dendritic trees is thought to influence dendritic computation, yet the relationship between branch-level structural properties and input spatial organization has not been systematically characterized. Here we analyze 1,234 dendritic branches across 12 electron-microscopy-reconstructed cortical neurons from the MICrONS dataset, classifying each branch into structural regimes by electrotonic length and quantifying spatial organization with three independent metrics. We find that spatial organization is systematically coupled to structural regime: electrotonically compact branches exhibit more spatially clustered synaptic inputs than expected under a geometry-matched null model (pairwise compactness z-score, mixed-effects $p = 9.3e-6$). Formal mediation analysis reveals that this coupling is largely driven by partner-level innervation structure -- compact branches receive concentrated input from fewer presynaptic partners making multiple contacts, while electrotonically isolated branches receive distributed input from many single-contact partners. The mediation is cell-type-specific: partner structure fully accounts for the coupling in excitatory neurons (100% mediated, bootstrap CI excludes zero) but only partially in inhibitory neurons (39% mediated), indicating a second, partner-independent organizational principle in inhibitory circuits. These results establish that electrotonic structure organizes synaptic input through cell-type-specific channels and provide an empirical foundation linking dendritic morphology to the spatial statistics of connectivity.

1 Introduction

Dendrites are not passive conduits. The spatial arrangement of synaptic inputs along dendritic branches can qualitatively change integration outcomes, enabling operations from linear summation to supralinear amplification depending on input proximity and dendritic properties (London & Hausser, 2005; Mel, 1994). Thin, electrotonically isolated branches can function as independent computational subunits (Polsky et al., 2004; Poirazi et al., 2003), while thick proximal dendrites sum inputs more linearly (Rall, 1967). This structural heterogeneity -- quantified by electrotonic length (L/λ) -- creates a landscape of potential computational modes along a single neuron's arbor (Koch, 1999).

If spatial organization of inputs were systematically coupled to structural regime, it would suggest that synaptic placement is not merely constrained by geometry but co-organized with the biophysical properties that determine how those inputs are integrated. Several lines of evidence hint at such coupling: functionally related inputs cluster on specific branches in sensory cortex (Jia et al., 2010; Druckmann et al., 2014), dendritic spikes require spatially clustered excitation (Larkum et al., 2009; Branco et al., 2010), and presynaptic partners concentrate their synapses on select branches (Kasthuri et al., 2015). However, whether these observations reflect a systematic relationship between branch structure and input spatial organization across diverse cell types has not been tested.

Here we address this question directly. Using serial electron microscopy reconstructions from the MICrONS dataset (MICrONS Consortium, 2021), we characterize the spatial organization of synaptic inputs on every branch of 12 cortical neurons (6 excitatory, 6 inhibitory including 2 bipolar cells). We classify branches into structural regimes by electrotonic length, quantify spatial organization with three independent metrics (Clark-Evans ratio, interval coefficient of variation, pairwise compactness), and test for coupling using mixed-effects models with rigorous confound controls.

We find that compact branches harbor more spatially clustered inputs than expected, while isolated branches approach uniform placement. Mediation analysis reveals that this coupling is largely explained by partner-level innervation architecture: compact branches attract repeat-contact partners whose boutons cluster, while isolated branches diversify across many single-contact partners. This mediation is complete in excitatory neurons but partial in inhibitory neurons, pointing to a second organizational axis in inhibitory circuits that remains to be characterized.

2 Results

2.1 Branch morphometry and structural regime classification

We computed branch-level features for 1,234 dendritic branches across 12 MICrONS cortical neurons (Table 1). Electrotonic length (L/λ) was calculated from cable theory using standard passive parameters ($R_m = 20$

kOhm*cm², $R_i = 150 \text{ Ohm*cm}$) applied to EM-measured branch lengths and diameters (Methods).

The observed electrotonic length distribution was compressed relative to textbook values: 62.4% of branches fell below $L/\lambda = 0.1$, the classical compact threshold (Rall, 1967), with a median of 0.059 and maximum of 0.50. This compression reflects the thicker diameters measured by EM compared to light microscopy, yielding larger space constants and smaller L/λ values. We therefore classified branches into structural regimes using tertile thresholds on the observed distribution ($L/\lambda < 0.027$: summation-like; $0.027 \leq L/\lambda < 0.117$: nonlinear-prone; $L/\lambda \geq 0.117$: compartmentalized), producing balanced groups of 411, 412, and 411 branches respectively (Figure 1).

Electrotonic regime classification was essentially independent of branch order (Cohen's kappa = -0.06), confirming that electrotonic length captures structural variation beyond simple topological depth. Sensitivity analysis across six alternative quantile thresholds showed stable coupling effect sizes, and the biophysical thresholds (0.1, 0.5) were included as a reference in all sensitivity comparisons.

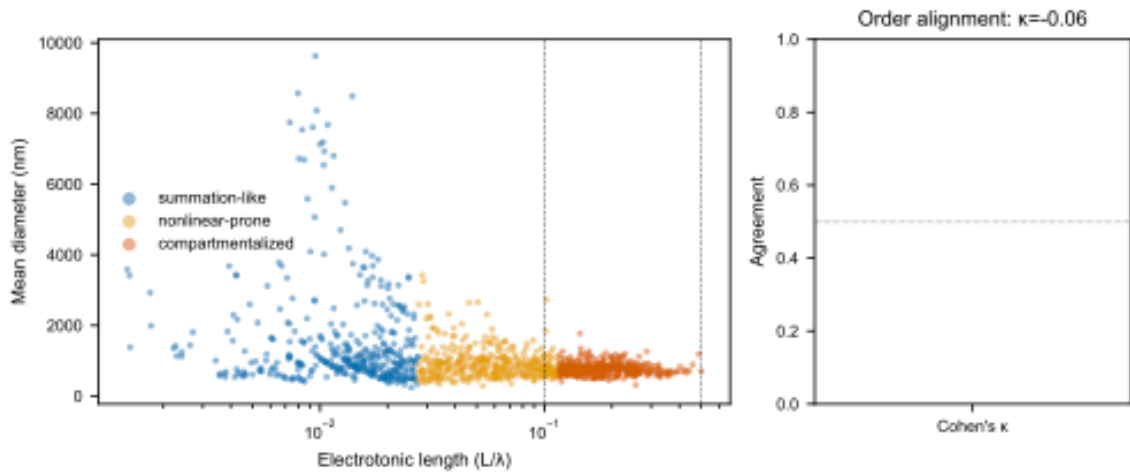


Figure 1. Structural regime classification by electrotonic length and diameter. Tertile thresholds on observed L/λ distribution produce balanced groups.

2.2 Spatial organization is coupled to structural regime

We quantified spatial organization of synaptic inputs on each branch using three independent metrics: Clark-Evans nearest-neighbor ratio, interval coefficient of variation (CV), and pairwise compactness (mean pairwise distance normalized by branch length). To control for the trivial dependence of raw metrics on branch geometry, we z-scored each metric against a per-branch uniform null (1,000 draws of k synapses on a branch of length L ; Methods).

Pairwise compactness showed the strongest coupling to regime (Figure 2). Summation-like branches had a mean z-score of -0.96 (substantially more compact than expected), while nonlinear-prone (-0.31) and compartmentalized (-0.39) branches were closer to the null expectation. A mixed-effects model with neuron as random intercept and synapse count, branch length, and excitatory fraction as covariates yielded significant regime coefficients (regime 1 vs. 0: $\beta = 0.60$, $p = 9.3e-6$; regime 2 vs. 0: $\beta = 0.82$, $p = 3.5e-5$). Permutation testing (10,000 regime-label shuffles within neurons) confirmed significance ($p < 0.001$).

Clark-Evans trended in the same direction (regime 2 coefficient: $\beta = 0.35$, $p = 0.047$; permutation $p = 0.002$), while interval CV showed a weaker pattern ($\beta = -0.41$, $p = 0.037$; permutation $p = 0.50$). The convergence of two metrics and the weakness of the third is informative: Clark-Evans and pairwise compactness both capture overall clustering tendency, while interval CV measures spacing regularity between consecutive synapses. Spatial organization by regime manifests as differences in overall clustering, not in the regularity of inter-synapse intervals.

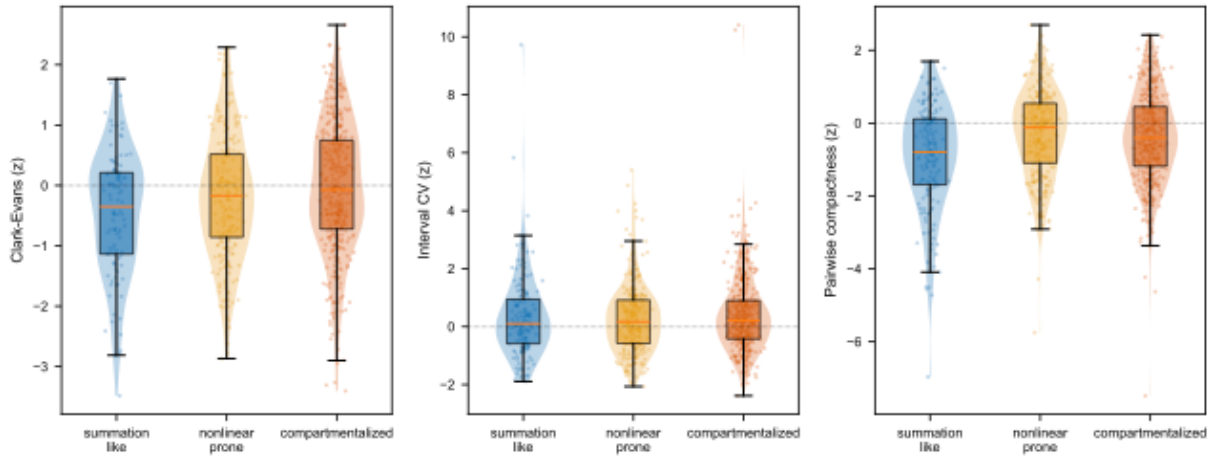


Figure 2. Spatial organization z-scores by structural regime for three independent metrics. Pairwise compactness shows the strongest coupling.

2.3 Confound controls

Three analyses addressed whether the coupling is an artifact of branch order or other confounds.

Residual analysis. Regressing pairwise compactness z-scores on branch order explained only 1.1% of variance ($R^2 = 0.011$). Kruskal-Wallis on the residuals by regime remained highly significant ($p = 1.2e-4$), confirming that regime captures structure beyond topological position.

Within-order stratification. Within individual branch-order strata, regime coupling persisted in the largest strata (order 1: $p = 0.0008$, $n = 97$; order 2: $p = 0.043$, $n = 148$). Of 18 testable strata, 3 reached significance at $p < 0.05$.

Sensitivity to thresholds. Coupling effect size was stable across all six quantile-based threshold variants (maximum absolute regime coefficient range: 0.81-1.27), confirming that the result is not an artifact of a particular threshold choice.

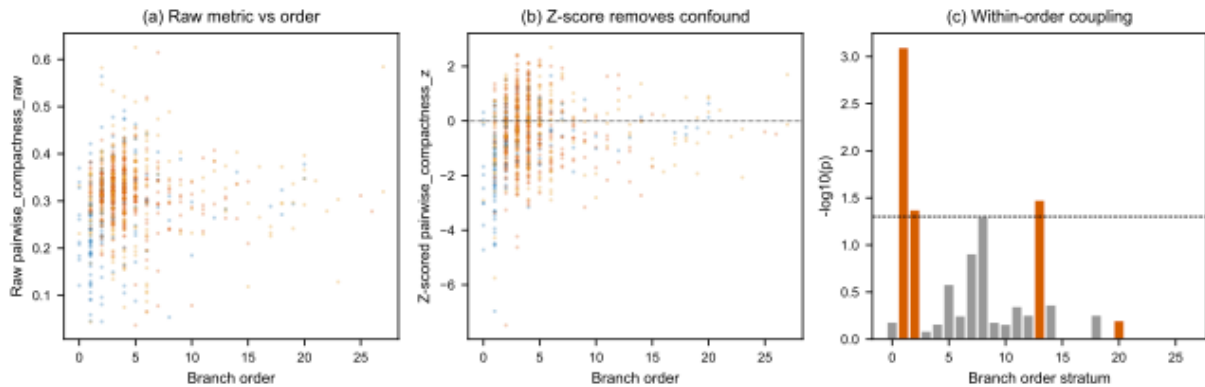


Figure 3. Confound controls. (a) Raw metric shows order confound, (b) z-scored removes it, (c) within-order stratified analysis.

2.4 Coverage bias assessment

The minimum synapse requirements for spatial metrics created regime-asymmetric exclusion: 61.6% of summation-like branches were excluded from pairwise compactness analysis (vs. 1.7% of compartmentalized branches), because compact branches are shorter and more likely to have few synapses. Excluded branches had a mean synapse count of 0.8 (vs. 16.5 for included branches). To test whether this selection drives the result, we verified that regime predicts partner structure even after controlling for synapse count ($p = 2.3e-43$), confirming that the coverage-surviving subset is not artifactually enriched.

2.5 Partner-level innervation structure mediates the coupling

The coupling between regime and spatial compactness could reflect either (a) a direct structural constraint on synapse

placement or (b) an indirect effect mediated by the partner-level innervation landscape. We found that compact branches receive qualitatively different innervation from isolated branches (Table 2): summation-like branches averaged 1.51 synapses per partner per branch with 30.5% of partners making multiple contacts, while compartmentalized branches averaged 1.19 synapses per partner with only 13.4% multi-contact partners. Conversely, compartmentalized branches received input from 19.7 unique partners on average versus 5.3 for summation-like branches.

Formal mediation analysis (Baron & Kenny, 1986) tested whether partner structure (mean synapses per partner per branch) mediates the regime-compactness link:

- Path a (regime $\rightarrow$ mediator): $a = -0.138$, $p = 1.1e-21$
- Path b (mediator $\rightarrow$ compactness, controlling regime): $b = -1.12$, $p = 5.0e-16$
- Direct effect after mediation: $c' = 0.069$, $p = 0.24$ (non-significant)
- Indirect effect: $ab = 0.156$; proportion mediated: 69.4%
- Sobel test: $z = 6.33$, $p = 2.4e-10$
- Bootstrap 95% CI for indirect effect: [0.093, 0.231] (5,000 resamples, excludes zero)

Synapse density provided a second mediation pathway (proportion mediated: 107%). In a joint model controlling both mediators simultaneously, the direct regime effect vanished entirely ($\beta = -0.06$, $p = 0.37$), while both density ($p = 5.4e-5$) and partner structure ($p = 1.4e-10$) remained significant. Mediation proportions exceeding 100% reflect suppression between correlated mediators sharing explained variance (density-partner $\rho = 0.34$; all variance inflation factors < 2), not overcounting of the indirect effect.

The coverage bias diagnostic confirmed that regime predicts partner structure even after controlling for synapse count (regime coefficients: $p = 1.3e-25$ and $p = 2.3e-43$), establishing that the partner gradient is genuinely tied to electrotonic structure rather than an artifact of differential branch inclusion.

2.6 Cell-type-specific mediation reveals two organizational axes

Excitatory neurons (493 branches, 6 neurons). The partner gradient was large (Cohen's $d = 0.89$, Mann-Whitney $p = 3.1e-11$). Pooled mediation was complete: total regime effect $c = 0.192$; direct effect after mediator $c' = -0.0004$; proportion mediated 100.2%. The spatial clustering on compact excitatory branches is entirely attributable to repeat-contact partners making clustered boutons.

Inhibitory neurons (277 branches, 4 non-BPC neurons). The partner gradient existed but with smaller effect size (Cohen's $d = 0.56$, significant at $p = 3.1e-9$ after controlling synapse count). Pooled mediation was partial: total effect $c = 0.260$; direct effect $c' = 0.159$; proportion mediated 38.9%. Over 60% of the regime-compactness coupling in inhibitory neurons operates through a partner-independent mechanism.

Bipolar cells (83 branches, 2 neurons). No significant partner gradient ($p = 0.07$ after controlling synapse count, Cohen's $d = 0.18$). Mediation was negligible (1.4%). The BPC coupling, though underpowered, appears independent of partner structure entirely.

The per-neuron gradient was consistent: 9 of 11 testable neurons showed higher synapses-per-partner on summation-like than compartmentalized branches (binomial $p = 0.033$). The two exceptions were both inhibitory (inh_MC, inh_BPC), consistent with the cell-type dissociation.

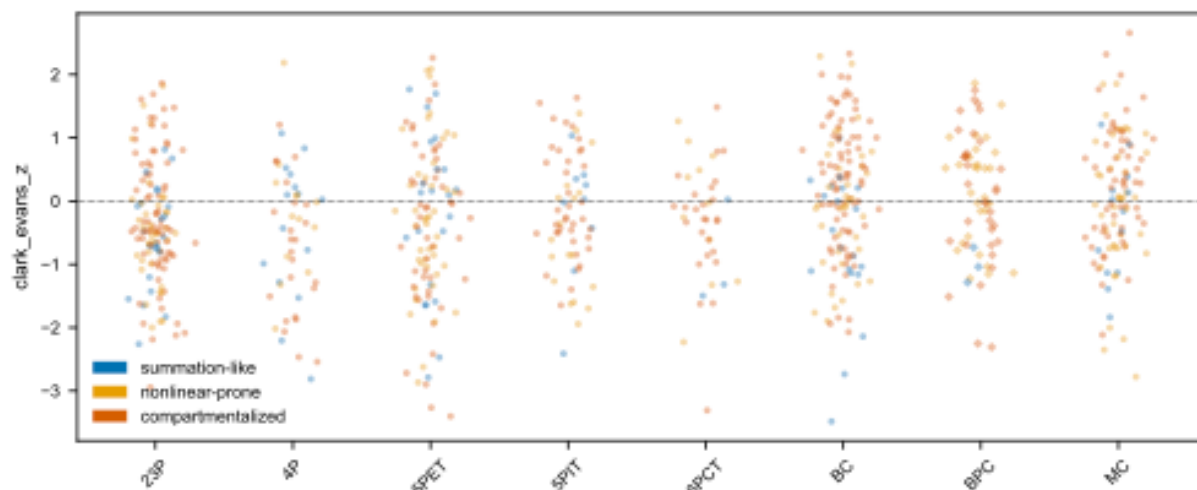


Figure 4. Cell-type-specific mediation. Excitatory neurons show complete mediation; inhibitory neurons show partial mediation with a residual partner-independent axis.

2.7 Partner compactness by regime

To connect these findings to partner-level spatial clustering, we mapped 1,254 presynaptic partners (each with ≥ 3 synapses) to the structural regime of their target branches. Partner compactness (observed mean pairwise distance / null median, "effect ratio") differed significantly by dominant target regime (Kruskal-Wallis $H = 9.46$, $p = 0.009$). Non-spanning partners (all synapses within one regime) showed greater compactness (median effect ratio 0.70) than regime-spanning partners (0.88), indicating that single-regime targeting is associated with tighter spatial organization.

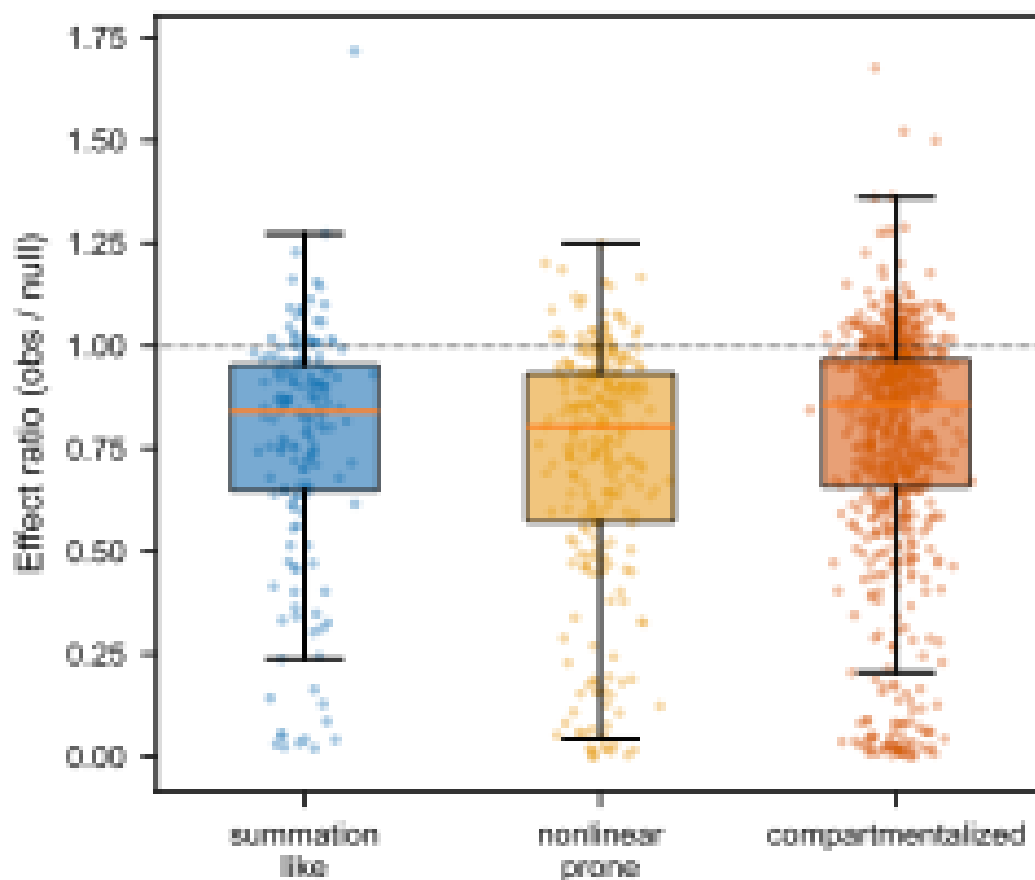


Figure 5. Partner compactness by structural regime of target branches.

3 Discussion

We report three principal findings. First, spatial organization of synaptic inputs is systematically coupled to dendritic structural regime across 1,234 branches from 12 cortical neurons. Second, this coupling is largely mediated by partner-level innervation architecture: compact branches consolidate input from repeat-contact partners whose boutons cluster, while isolated branches diversify across many single-contact partners. Third, the mediation is cell-type-specific, revealing two distinct organizational axes -- one fully explained by partner structure in excitatory neurons, and a second, partner-independent axis in inhibitory neurons.

3.1 The direction of coupling

The observed coupling -- compact branches more clustered, isolated branches closer to random -- is the opposite of what a simple nonlinear-subunit model would predict (where isolated branches would need clustered inputs for local spike generation). This does not contradict the subunit hypothesis but rather suggests that spatial organization is not optimized for computation at the single-branch level. Instead, it reflects the innervation constraints: many axons converging on a short, accessible proximal branch create clustering as a byproduct of repeated contact, while the longer path to distal branches favors single-contact diversification.

3.2 Two organizational axes

The cell-type dissociation is the most generative finding. In excitatory neurons, partner structure fully explains the coupling -- compact branches are clustered because they attract repeat visitors. In inhibitory neurons, a substantial residual (61%) persists after accounting for partner structure, pointing to a second organizational principle. This could reflect zone-specific inhibitory targeting logic, where different classes of inhibitory afferents (e.g., basket cells vs. chandelier cells vs. Martinotti cells) preferentially target specific dendritic compartments regardless of how many synapses each makes. The precise characterization of this partner-independent axis remains an open question.

3.3 Electrotonic length distribution

The finding that 62% of MICrONS branches fall below $L/\lambda = 0.1$ warrants comment. EM-reconstructed dendrites yield thicker diameter measurements than light microscopy, inflating the space constant λ and compressing L/λ . Our tertile thresholds (0.027, 0.117) capture relative structural variation within this compressed range. The coupling we report is a graded relationship that is stable across threshold choices (sensitivity analysis), not a sharp transition between discrete computational modes. This is consistent with dendritic biophysics: electrotonic properties vary continuously, and integration mode depends on the interaction of structure, channel expression, and input timing (Stuart et al., 1997).

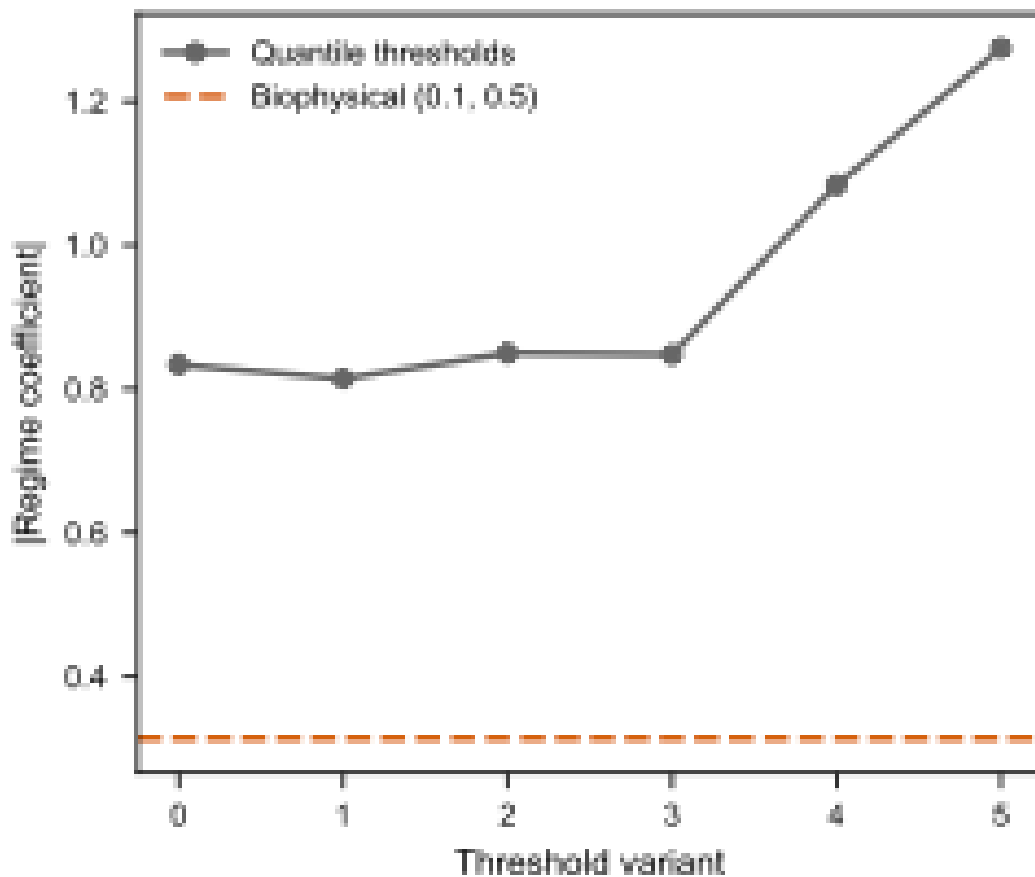


Figure 6. Sensitivity analysis: coupling effect size across threshold choices.

3.4 Limitations

The coverage bias from minimum synapse requirements creates asymmetric exclusion (61.6% of summation-like branches excluded vs. 1.7% of compartmentalized branches). Although the coverage diagnostic confirms the partner gradient survives controlling for synapse count, the summation-like branches that contribute to our analysis are a filtered subset with atypically high synapse density. Per-neuron mediation estimates are variable due to limited within-neuron sample sizes, though the pooled mediation is robust (bootstrap CI excludes zero). Passive cable parameters were assumed throughout; active conductances would modify the effective electrotonic structure.

4 Methods

4.1 Data

Twelve neurons (6 excitatory: 23P, 23P, 4P, 5P-ET, 5P-IT, 6P-CT; 6 inhibitory: 2 basket cells, 2 Martinotti cells, 2 bipolar cells) were selected from the MICrONS cortical mm³ dataset (MICrONS Consortium, 2021). SWC skeleton files and synapse coordinates were obtained from prior reconstruction. Presynaptic partner identity and broad cell type (excitatory/inhibitory) were determined from the connectome.

4.2 Branch morphometry

Skeletons were loaded with a scale factor of 1,000 (converting μm to nm) and filtered to dendritic compartments (SWC types 1, 3, 4). Synapses were snapped to the skeleton using a KD-tree nearest-neighbor search with a maximum distance threshold of 50,000 nm. Branches were defined as maximal paths between branching or terminal points.

For each branch, we computed: branch order (BFS from soma), mean diameter ($2 \times \text{mean SWC radius}$), total path length, electrotonic length (L/λ , where $\lambda = \sqrt{R_m d / 4 R_i}$, $R_m = 20,000 \text{ Ohm}\cdot\text{cm}^2$, $R_i = 150 \text{ Ohm}\cdot\text{cm}$, $d =$

mean diameter), soma distance (geodesic via Dijkstra on branch endpoints), synapse count, and excitatory/inhibitory synapse counts from partner cell type classification.

4.3 Regime classification

Branches were classified into three structural regimes by electrotonic length using tertile thresholds computed from the observed distribution. The 33rd percentile ($L/\lambda = 0.027$) separated summation-like from nonlinear-prone; the 67th percentile ($L/\lambda = 0.117$) separated nonlinear-prone from compartmentalized. Sensitivity analysis swept six alternative quantile-based threshold pairs and included the biophysical reference thresholds (0.1, 0.5).

4.4 Spatial organization metrics

Clark-Evans ratio (≥ 5 synapses). Mean observed nearest-neighbor distance divided by expected under uniform placement with Donnelly edge correction (Clark & Evans, 1954; Donnelly, 1978): $E[NN] = L/(2n) + 0.0514 \cdot L/n^2$. Values below 1 indicate clustering.

Interval CV (≥ 3 synapses). Standard deviation of inter-synapse gaps divided by their mean. Values above 1 indicate clustered (heterogeneous) spacing.

Pairwise compactness (≥ 3 synapses). Mean pairwise distance between synapses divided by branch length. Lower values indicate more compact distributions.

4.5 Branch null model and z-scoring

For each branch with k synapses on length L , 1,000 draws of k positions from $\text{Uniform}(0, L)$ generated null distributions for each metric. The z-score for each branch was $(x_{\text{obs}} - \mu_{\text{null}}) / \sigma_{\text{null}}$. This controls for branch geometry (length and synapse count) so that z-scores are comparable across branches.

4.6 Statistical testing

Mixed-effects model. $z\text{-score} \sim C(\text{regime}) + \text{synapse_count} + \text{branch_length} + \text{exc_fraction} + (1|\text{neuron})$ via REML. Neuron as random intercept pools information across the 12 neurons while accounting for between-neuron variability.

Permutation test. Regime labels were shuffled within each neuron (10,000 permutations). The test statistic was the difference in mean z-score between regime 0 and regime 2.

Per-neuron Kruskal-Wallis. Nonparametric test of z-score by regime within each neuron, corrected for multiple comparisons via Benjamini-Hochberg FDR (Benjamini & Hochberg, 1995).

Within-order stratification. Kruskal-Wallis on z-score by regime within each branch-order stratum. If coupling persists within order strata, it is not an artifact of branch order.

Residual analysis. Regressed z-score on branch order via OLS, then tested regime effect on residuals.

4.7 Mediation analysis

Mediation followed the Baron and Kenny framework (1986) with bootstrap inference. For each candidate mediator M (mean synapses per partner per branch; synapse density): Path a: regime $\rightarrow M$ (OLS); Path b: $M \rightarrow \text{compactness } z$, controlling regime (OLS); Indirect effect: ab ; proportion mediated: $(c - c')/c$. Statistical significance was assessed via the Sobel test (Sobel, 1982) and bootstrap resampling (5,000 iterations, percentile confidence intervals). Mediation proportions exceeding 100% indicate suppression between correlated mediators sharing explained variance (confirmed by variance inflation factors < 2 for all predictors).

4.8 Software

All analyses were implemented in Python using NumPy, SciPy, pandas, statsmodels, scikit-learn, and the Neurostat package for dendritic spatial statistics. Code is available at <https://github.com/saivoru777-ship-it/dendritic-regime-coupling>.

Tables

Table 1. Summary of neurons analyzed.

Label	Cell type	Branches	Synapses	Median L/lambda
exc_23P	Excitatory (L2/3)	132	1,063	0.057
exc_23P_2	Excitatory (L2/3)	122	937	0.047
exc_4P	Excitatory (L4)	84	990	0.064
exc_5PET	Excitatory (L5 ET)	220	2,490	0.043
exc_5PIT	Excitatory (L5 IT)	141	1,199	0.062
exc_6PCT	Excitatory (L6 CT)	102	488	0.063
inh_BC	Basket cell	91	1,617	0.106
inh_BC_2	Basket cell	100	1,491	0.065
inh_MC	Martinotti cell	79	1,584	0.047
inh_MC_2	Martinotti cell	59	1,230	0.072
inh_BPC	Bipolar cell	56	669	0.072
inh_BPC_2	Bipolar cell	48	586	0.060
Total		1,234	14,344	0.059

Table 2. Partner innervation structure by structural regime.

	Summation	Nonlinear	Compartment.
Mean synapses/partner	1.51	1.22	1.19
Frac. multi-contact	30.5%	16.0%	13.4%
Frac. syn from multi	44.5%	26.4%	24.6%
Mean partners/branch	5.3	9.4	19.7

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Supplementary Figures

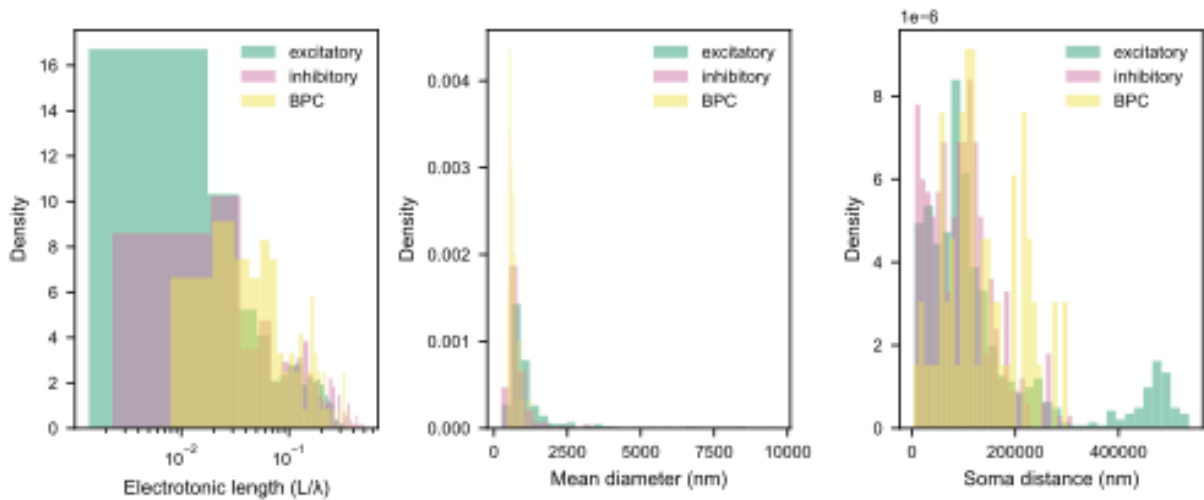


Figure S1. Distributions of electrotonic length, diameter, and soma distance across all 1,234 branches, colored by neuron type.

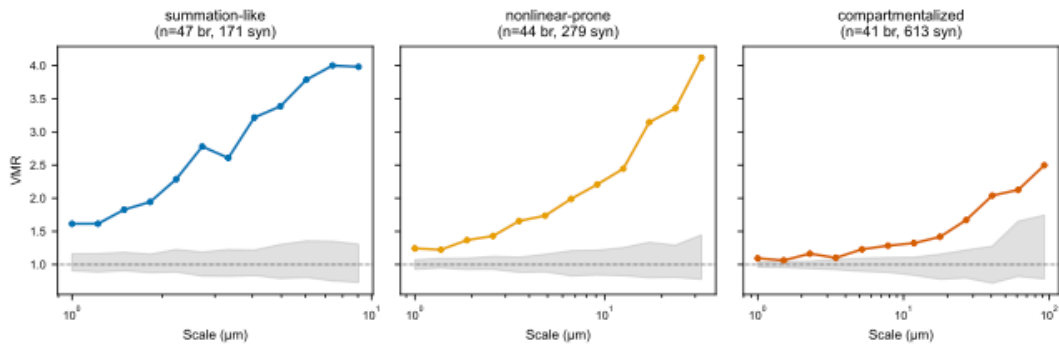


Figure S2. Variance-mean ratio fingerprint curves per structural regime with null envelopes.

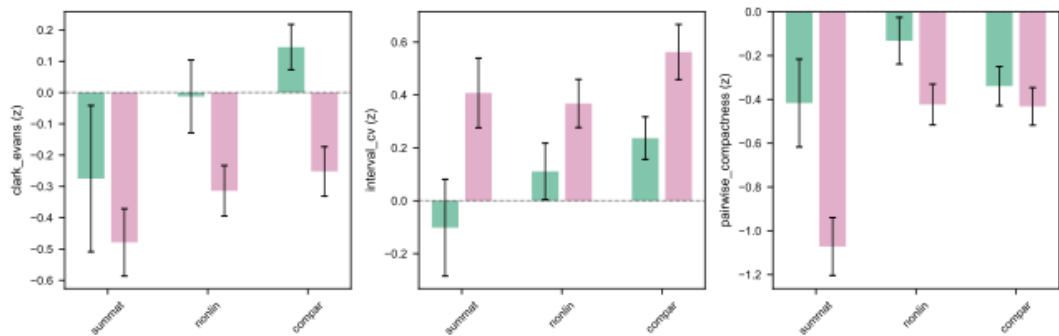


Figure S3. Excitatory vs. inhibitory spatial organization within each regime.